

CHAPTER 2 Guide for Certification of Cranes

SECTION 5 Loose Gear Items, Sheaves, and Special Components (1 August 2024)

1 Objective (1 August 2024)

1.1 Goals

Loose gear items, sheaves, and special components covered in this section are to be designed, constructed, certified and maintained to:

<i>Goal No.</i>	<i>Goal</i>
OTH SG23	provide means for the safe lifting, handling or transfer of cargo or personnel.
OTH SG25	prevent failures or accidents, which would cause harm to personnel or damage to ships, offshore facilities, subsea or shore-based structures, as well as the environment.

The goals in the cross-referenced sections of this document or other ABS Rules are also to be met.

1.2 Functional Requirements

In order to achieve the above stated goals, the design, construction, installation and maintenance of the loose gear items, sheaves and special components are to meet the following functional requirements:

<i>Functional Requirement No.</i>	<i>Functional Requirements</i>
Other Systems (OTH)	
OTH-FR1	Loose gear items, sheaves, and special components are to have sufficient strength and integrity throughout their service life, when the crane is operated within its design and operational parameters.
OTH-FR2	Sheaves are to guide the ropes appropriately for their intended application, and are to prevent rope damage (<i>e.g., due to abrasion, pinching or bird-caging</i>).
OTH-FR3	Sheaves are to be provided with means to retain the ropes within the sheave grooves, and are to prevent the ropes from slipping.

<i>Functional Requirement No.</i>	<i>Functional Requirements</i>
OTH-FR4	Means are to be provided to lubricate the sheave bearings, so as to reduce friction and wear.
OTH-FR5	Hooks are to be provided with means to retain the rigging on the hook, so as to prevent accidental dropping of the load.
OTH-FR6	Loose gear items and special components are to be permanently marked with their SWL and weight so as to prevent their inadvertent use for non-rated loads.
OTH-FR7	The sheave pitch diameter to rope diameter ratio is to be adequate for the intended application, so as to minimize rope fatigue.

The functional requirements in the cross-referenced sections of this document or other ABS Rules are also to be met.

1.3 Compliance

Loose gear, sheaves and special components are considered to comply with the Goals and Functional requirements within their scope of certification when the applicable prescriptive requirements are complied with or when an alternative arrangement has been approved. Refer to Part 1D Chapter 2.

2 Loose Gear

2.1 General (1 August 2024)

Loose gear items (such as chains, rings, hooks, hook blocks, links, shackles, swivels, spreader bars/beams, strongbacks, and grabs) are to be tested with a proof load at least equal to the following:

<i>Article of Gear</i>	<i>Proof Load</i>
Chains, rings, hooks, links, shackles, swivels, etc. SWL ≤ 25 t SWL > 25 t	2 x SWL (1.22 x SWL) + 20
Single sheave block	4 x SWL
Multi-sheave blocks and hook blocks SWL ≤ 25 t 25 t < SWL ≤ 160 t 160 t < SWL	2 x SWL (0.993 x SWL) + 27 1.1 x SWL
Lifting beams, spreader beams, frames, grabs SWL ≤ 10 t 10 t < SWL ≤ 160 t 160 t < SWL	2 x SWL (1.04 x SWL) + 9.6 1.1 x SWL

Notes:

- 1 Safe working load (SWL) of the loose gear is the maximum static load that an item of loose gear is certified to lift.
- 2 Sheave blocks that are permanently attached to, or are integral part of the hook, are called hook blocks. Hook blocks are to be tested with the load for multi-sheave blocks. The hook of the hook block is to be tested with the loads for hooks.

- 3 The SWL for a single sheave block, including single sheave blocks with becketts, is to be taken as one half of the resultant load on the head fitting.
- 4 The SWL for a multi-sheave block is to be taken as the resultant load on the head fitting.
- 5 *(1 September 2012)* Alternatively, the proof tests as recommended in the latest applicable edition of the I.L.O. publication “Code of Practice on Safety and Health in Port” may be accepted where the items of gear are manufactured or tested or both to the requirements of those regulations and are intended for use on vessels under jurisdictions accepting them.
- 6 The safe working load to be marked on a single sheave block is to be the maximum load which can safely be lifted by the hook suspended from the body of the block.

Evidence of compliance with the proof load test requirements in this Section for all rings, hooks, links, shackles, swivels, blocks, and any other loose gear whether accessory to a machine or not, but which is used as crane gear is to be listed on an appropriate certificate as required by [1.3].

Loose gear are to undergo NDE after proof load testing in accordance with a recognized standard, such as DOE STD 1090, ASME B30.10, ASTM E709 (MT), and ASTM E165 (PT), by the loose gear manufacturer. Results are to be made available to the Surveyor upon request.

Structure, loose gear, and/or containers used solely for shipping or transferring equipment to offshore units are not subject to the requirements of this Section. The ABS *Guide for the Certification of Offshore Containers* may be applied for these items outside the scope of this document.

2.3 Certificates

Articles of loose gear are to have a certificate furnished by the manufacturer. The certificate is to show the distinguishing number or mark applied to the article of gear, description, kind of material, carbon content, date of test, proof load applied, and safe working load. Loose gear certificates are to be inserted in the Register of Lifting Appliances. See [2-8/1]. The safe working load SWL is to be marked on the hoist blocks.

2.5 Special Components *(1 August 2024)*

Blocks of special nature, together with their connecting components, spreader beams, strongbacks and other special lifting devices and components built into or for cranes, heavy lift gear, crane hooks or hoisting machinery which are specially designed for use with a particular lifting unit, the designs of which are submitted for approval as steel structural parts (including hook curves indicating the hook load reductions based on sling angle and eccentricity, as applicable), in accordance with Section [2-2], need not be considered loose gear for the purpose of certification. For material requirements, see [2-3/1]. Surveys during construction at the component manufacturer’s plant are to be carried out in accordance with [2-7/3]; testing and examination are to be carried out with the gear as a unit, as required by [2-7/5].

For crane hooks, appropriate nondestructive examination, in accordance with a recognized standard, such as DOE STD 1090, ASME B30.10, ASTM E709 (MT), and ASTM E165 (PT), is to be performed after proof load testing to the satisfaction of the attending Surveyor.

Nondestructive examination will also be required for other components where visual inspection is considered to be inadequate.

3 Sheaves *(1 December 2024)*

- i) Sheaves grooves are to be smooth and free from surface defects which could cause rope damage. The cross-sectional radius at the bottom of the groove is to be such so as to form a saddle for the size of rope used; the sides of the groove are to be tapered outwardly to facilitate entrance of the rope into the groove. Flange corners are to be rounded and the rims are to run true about the axis of rotation.
- ii) All sheaves including running blocks are to be provided with guards or other suitable devices to prevent the rope from coming out of the sheave groove.
- iii) Means are to be provided, if necessary, to prevent chafing of the ropes.

- iv) All sheave bearings are to be provided with means for lubrication. Permanently lubricated bearing are exempt from this requirement.
- v) Sheave pitch diameter to rope diameter ratio for crane running wire ropes is not to be less than 18, or 20 for sheaves used in motion compensation systems, and for standing rigging wire ropes is not to be less than 10.
- vi) The sheave groove inside diameter is to be larger than the rope diameter as per the following:
 - For Steel sheaves, 6% - 10%,
 - For Cast nylon sheaves, 3% - 27%Other sheave groove diameters will be subject to ABS design review and approval.

Commentary:

For additional guidance on the sheave groove inside diameter, refer to the wire rope manufacturer's instructions.

End of Commentary

- vii) Where sheaves are built into the structure of the crane need not be tested and certified as loose gear, but will be accepted based on verification of compliance with the requirements of to above, material verification in accordance with Section [2-3] and testing and examination with the gear as a unit in accordance with [2-7/5]. Materials are to be as required in [2-3/5.11].
- viii) Sheaves of special nature which are specially designed for use with a particular lifting unit, the designs of which are submitted for approval as steel structural parts, are to be treated as special components in accordance with [1.5]. For sheaves made of polymer materials known as Type 6 cast nylons, the allowable stresses for bending, shear, compression, bearing etc. are to be limited to 30% of the corresponding material strength for bending, shear, compression, bearing, etc. Materials are to be as required in [2-3/5.11].

5 Hooks and Hook Blocks

5.1 General (1 August 2024)

Hook blocks are to be of sufficient weight to overhaul the line from the highest hook position giving consideration to the boom length, jib length, as well as the number of parts of line in use.

Hook blocks are to be clearly and permanently marked with their maximum rated capacity (SWL) and weight.

Hooks are to be equipped with latches, unless equivalent means are provided to retain the rigging on the hook. When provided, the latch is to bridge the throat opening of the hook for the purpose of retaining slings or other lifting devices, under slack conditions.

When hooks and hook blocks are used for personnel lifting, they are to comply with the additional requirements of [2-9/5].

5.3 Shipboard Cranes

Main and auxiliary hook blocks and hooks for shipboard cranes may be accepted as loose gear on the basis of testing and manufacturer certification as per [1.1] and [1.3]. Alternatively, they are to be certified as "Special Components" as per [1.5].

When hooks and hook blocks are certified as "Special Components", the design load is to be calculated as per [2-2/7]. The calculated stresses are to be in compliance with [2-2/5].